

Downloading and Installing VMware

Before we show you how to install Kali Linux on VMware, you need to have VMware installed on your system. There are two types of VMware Workstation available, including a free and a paid version.

Free Version (Player)

The free version of VMware is called **VMware Workstation Player**. It is a basic version of the software that is suitable for personal use. With VMware Workstation Player, you can create and run virtual machines on your desktop or laptop. It supports a wide range of operating systems, including Windows, Linux, and macOS. However, it has limited features compared to the paid version.

Paid Version (Pro):

The paid version of VMware is called **VMware Workstation Pro**. It is a more advanced version of the software that is suitable for professional use. With VMware Workstation Pro, you can create and run virtual machines on your desktop or laptop, as well as on servers. It also supports the same operating systems as the free version.

Additionally, it has advanced features, such as the ability to create snapshots, clone a VM, and change the virtual network settings.

At the time of writing, VMware Workstation Pro retails for \$199 USD.

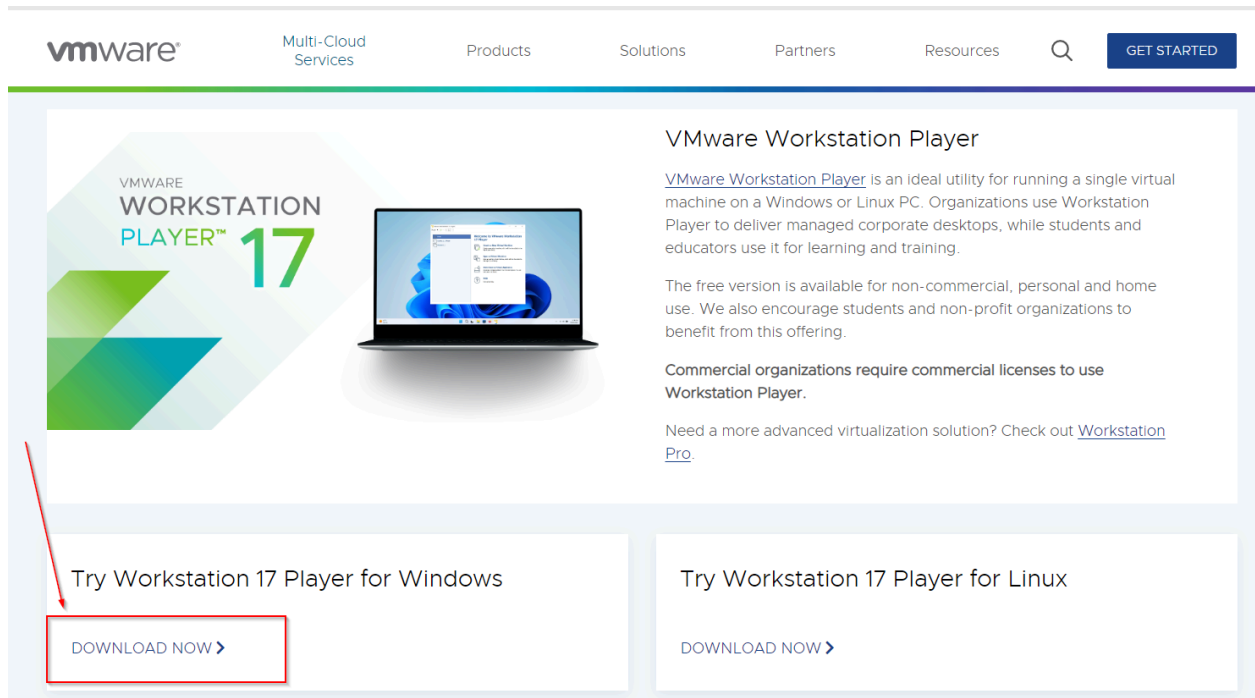
VMware also has an **academic discount** which will cost you \$119 USD. To take advantage of this, you need to be a student, faculty member, staff member or administrator at an educational institution, such as a

college, university, or public or private K-12 school. Additionally, parents or legal guardians purchasing on behalf of a child who is a student also qualify.

To download and install VMware, follow these steps:

Go to the **VMware Workstation Player** or **VMware Workstation Pro website** (also linked above).

Click the "Download Now" button for the version you want to download. The installation process is identical for both the Player and Pro version.



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VMware Workstation Player

VMware Workstation Player is an ideal utility for running a single virtual machine on a Windows or Linux PC. Organizations use Workstation Player to deliver managed corporate desktops, while students and educators use it for learning and training.

The free version is available for non-commercial, personal and home use. We also encourage students and non-profit organizations to benefit from this offering.

Commercial organizations require commercial licenses to use Workstation Player.

Need a more advanced virtualization solution? Check out [Workstation Pro](#).

Try Workstation 17 Player for Windows

DOWNLOAD NOW >

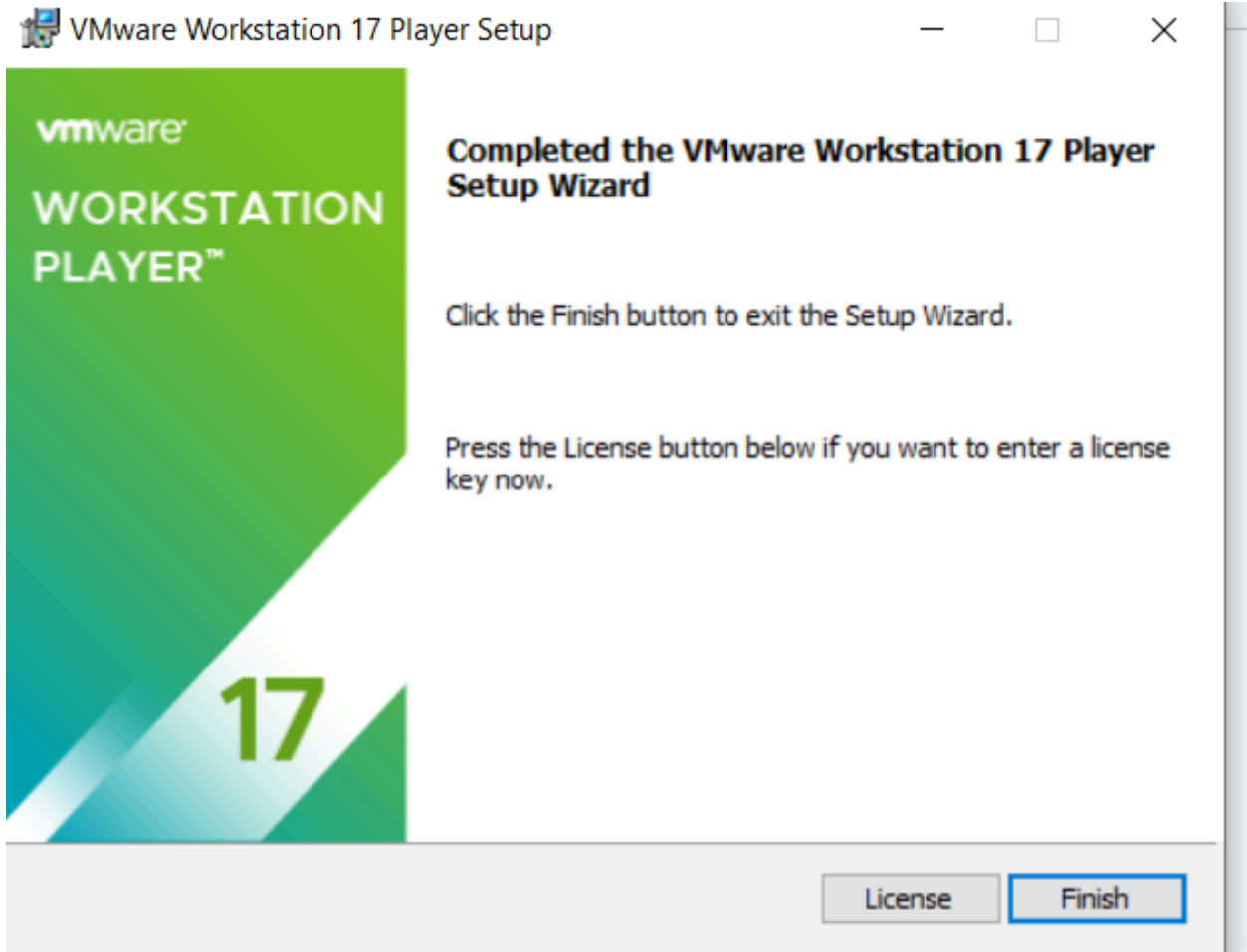
Try Workstation 17 Player for Linux

DOWNLOAD NOW >

Download the installer, and when completed, double-click on the VMware installer file from the folder you saved it to and start the installation process. Click Next.



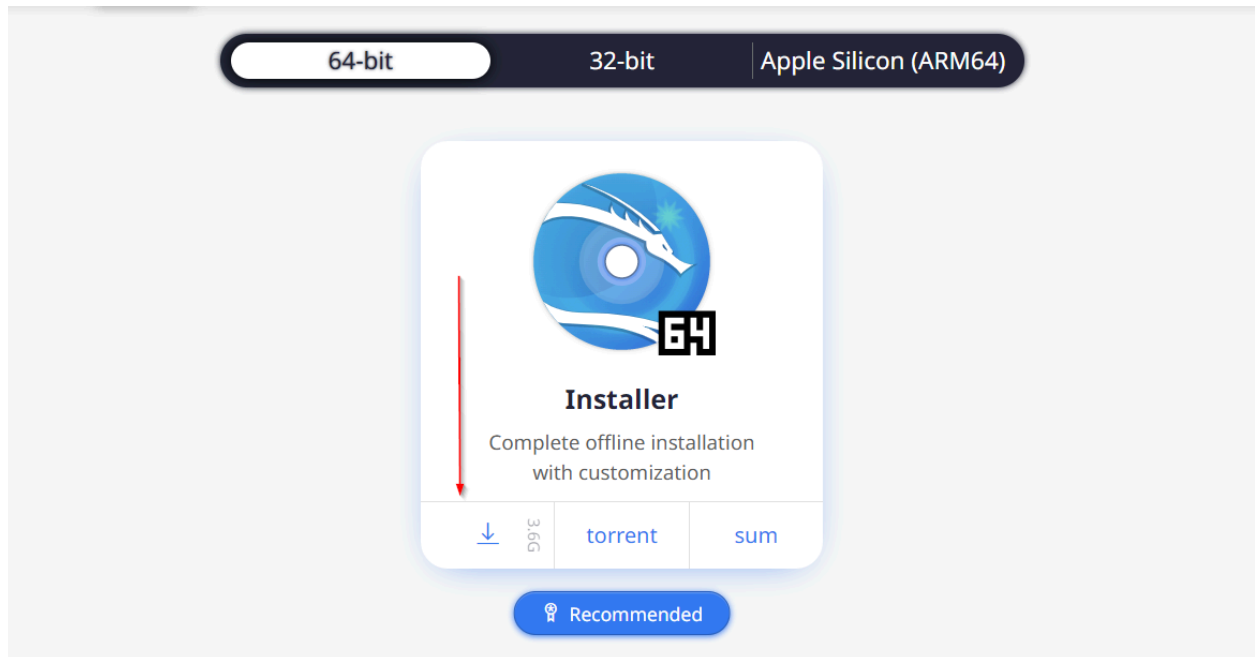
Accept the license agreement and click Next. Click Next three more times, and finally, click Install. Once VMware is installed, click Finish, and you can proceed to install Kali on VMware.



Downloading Kali

There are two ways we can download and install Kali on VMware, using the .iso or the VMware image file. We recommend using the prebuilt VMware file, which is completely configured and ready to go. It's a much easier process and takes little time to get it up and running.)

Alternatively, if you want to download the .iso file, head over to the Kali website and download the [Kali installer image](#).

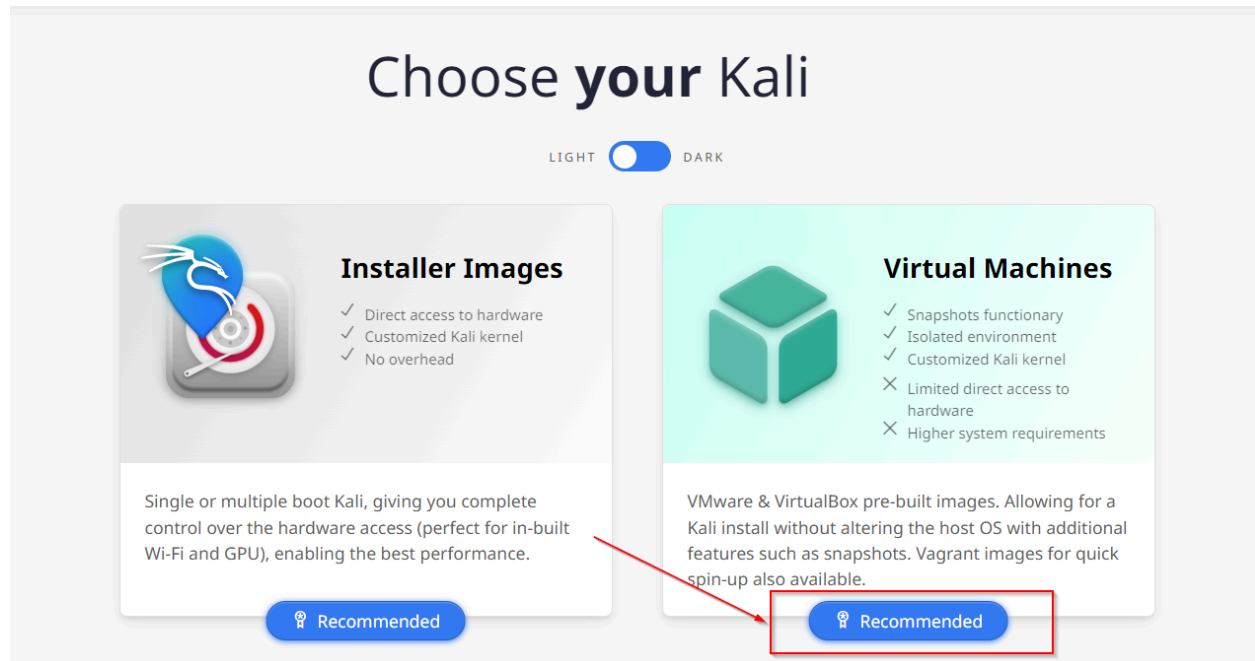


We will walk through setting up the .iso file in a section below.

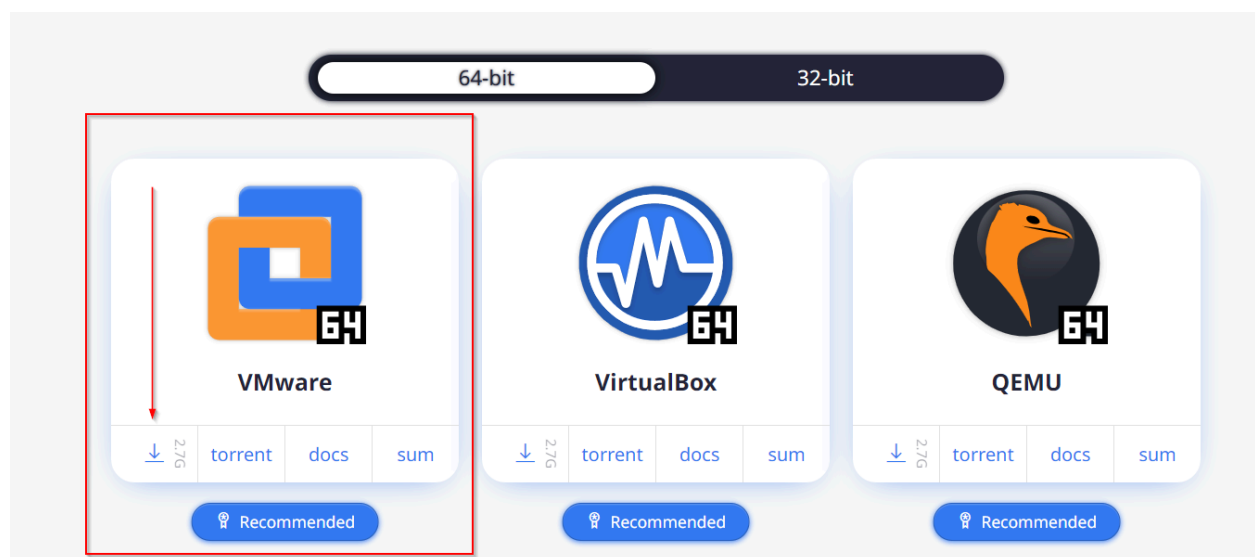
Using a Virtual Machine Image

To download the Kali VMware image, follow these steps:

1. Go to the official Kali Linux website ([**Download Kali**](#)).
2. Scroll down to the "Virtual Machines" section.



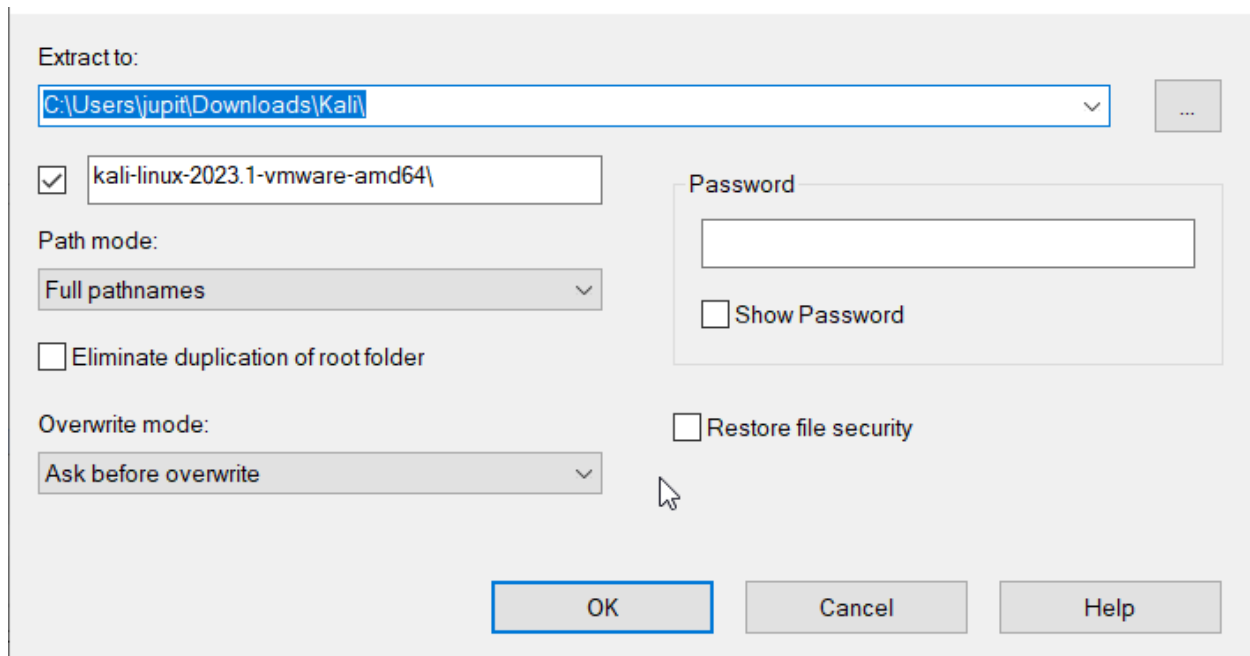
3. Choose the VMware platform and click on the download link.



How to Install Kali Linux on VMware

Once the download is complete, you will notice it saved as a .zip file. Double-click on the downloaded file to open it with your default zip program. If you need a zip program, we recommend [7Zip](#).

Set where you would like the file to be extracted and click OK.



We will show you how to Install the image on both the Player and Pro versions, which are very similar.

VMware Workstation Player

Open VMware Workstation Player and click on “Open a Virtual Machine.”

 Home

Welcome to VMware Workstation 17 Player



Create a New Virtual Machine

Create a new virtual machine, which will then be added to the top of your library.



Open a Virtual Machine

Open an existing virtual machine, which will then be added to the top of your library.



Download a Virtual Appliance

Download a virtual appliance from the marketplace. You can then open it in Player.



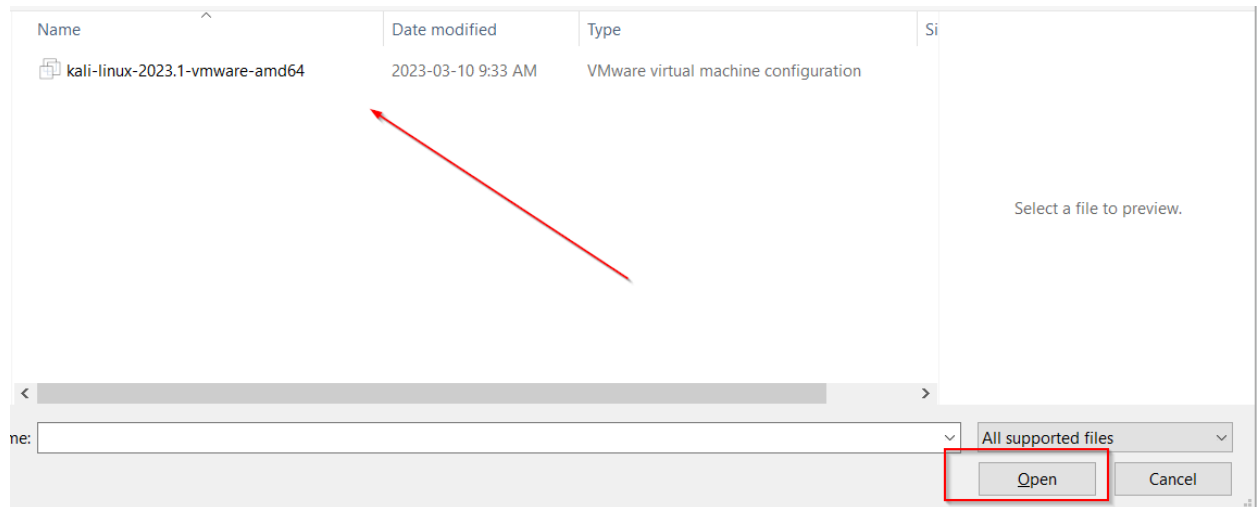
Help

View online help.

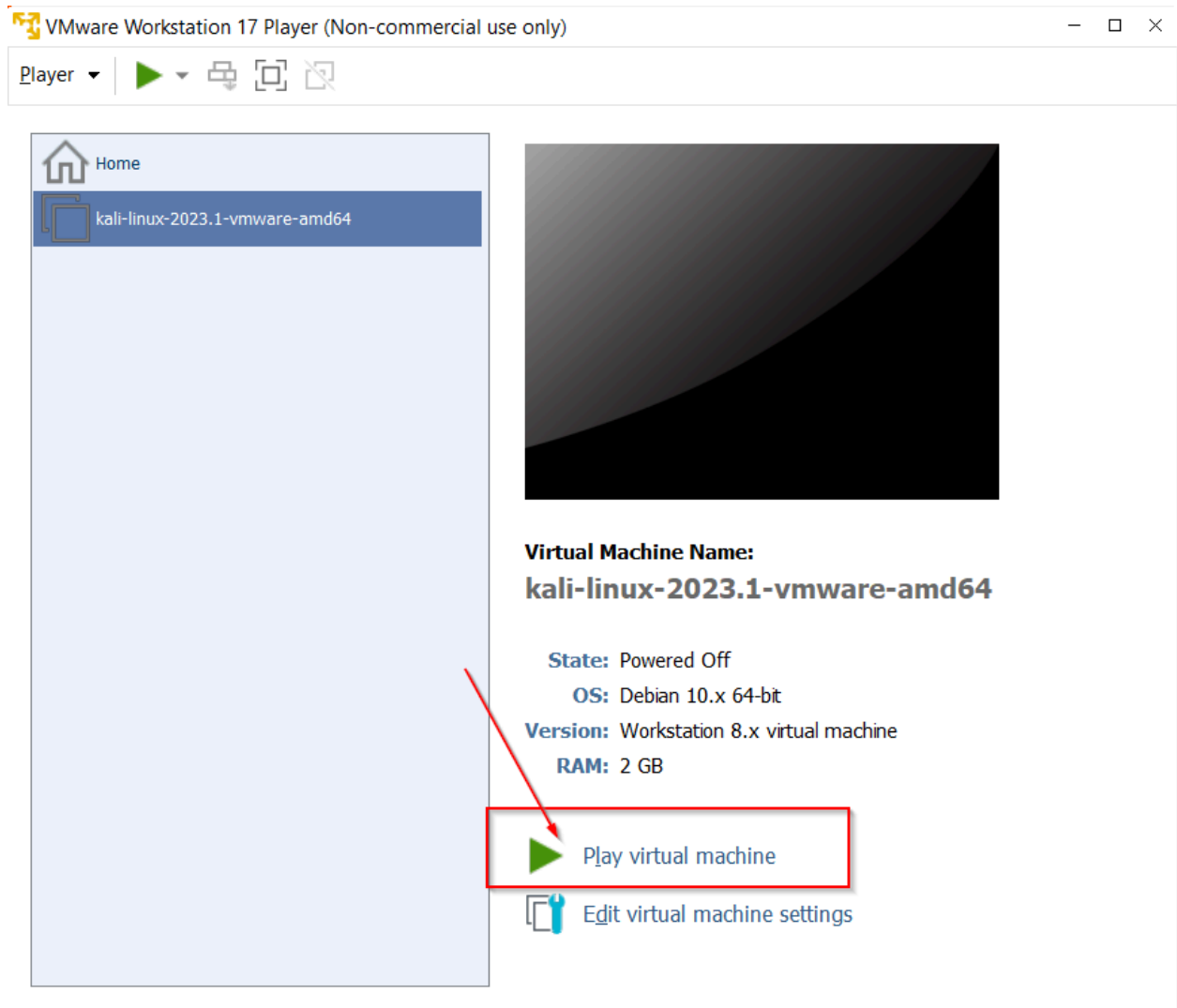


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Locate the folder where you extracted the file and click on it. The file you are looking for is called VMware virtual configuration file. Click Open.

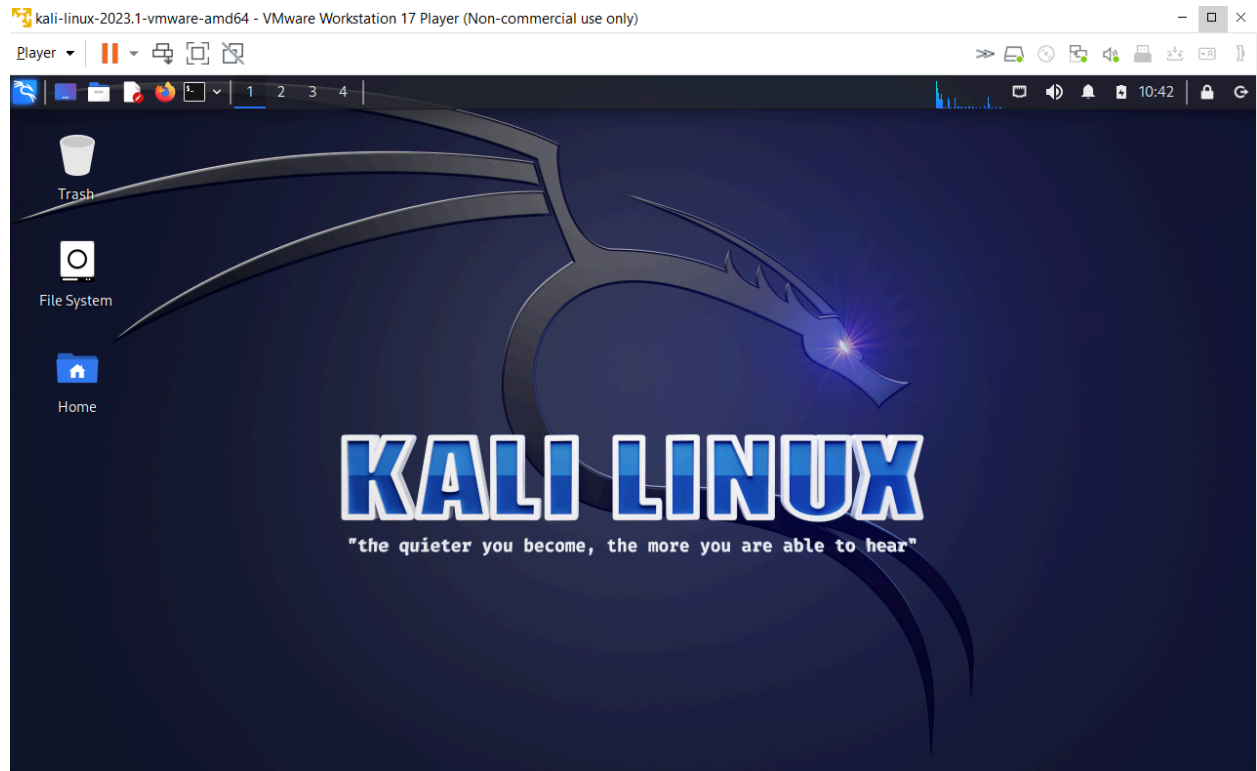


And just like that, the Kali image file is ready to go. To start Kali, click on “Play virtual machine.”



The default login credentials are

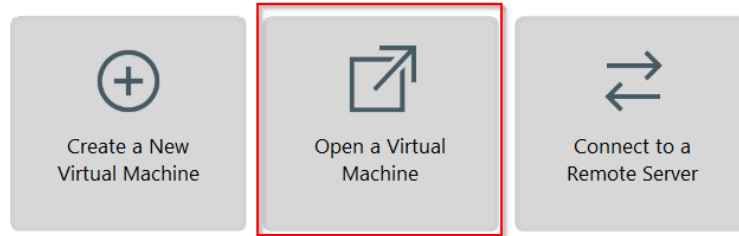
- Username kali
- Password kali



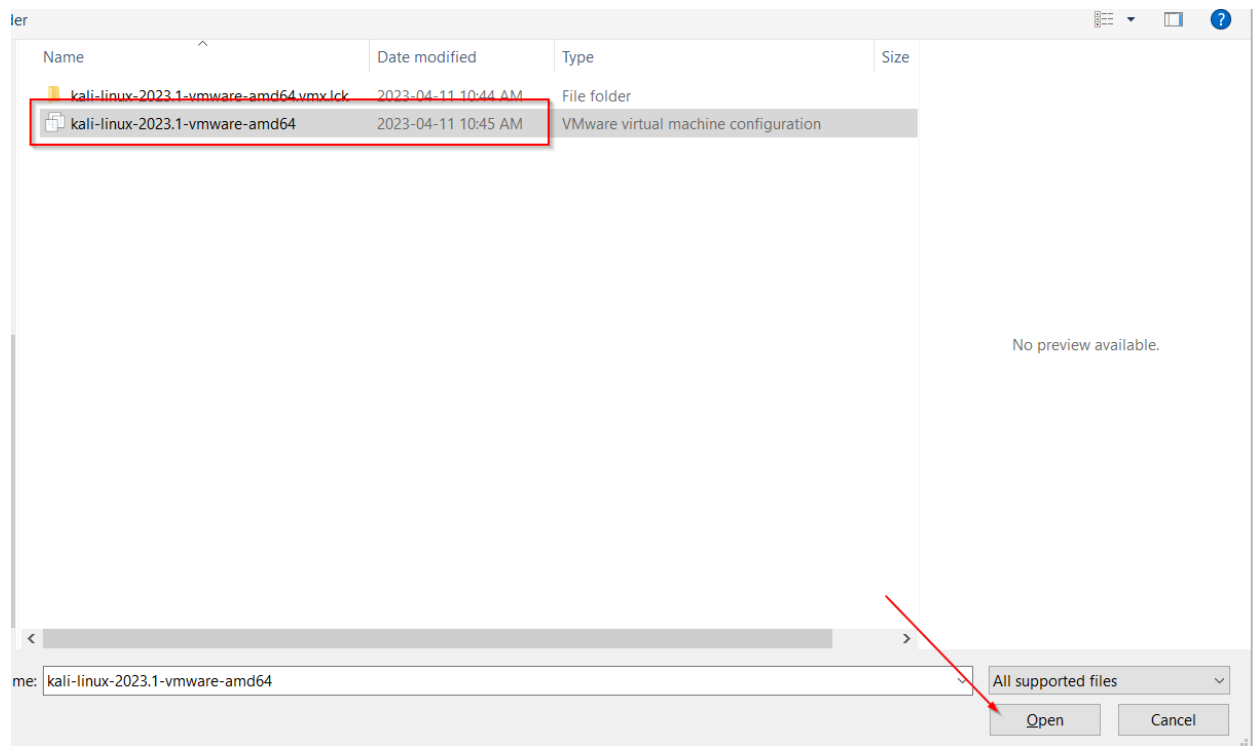
VMware Workstation Pro

Here is how to install Kali Linux on VMware Workstation Pro, which is very similar to how we installed it on VMware Workstation Player. Open VMware Pro and click on “Open a Virtual Machine.”

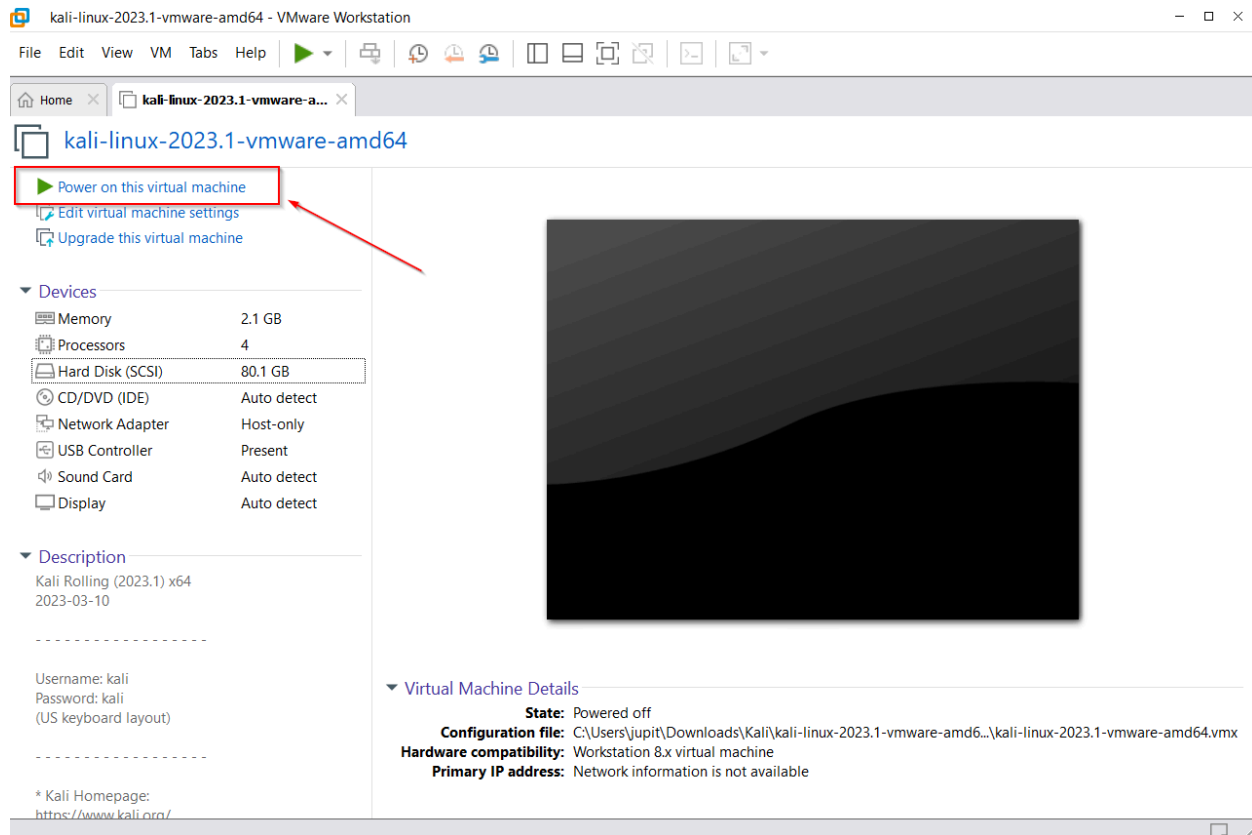
WORKSTATION PRO™ 17

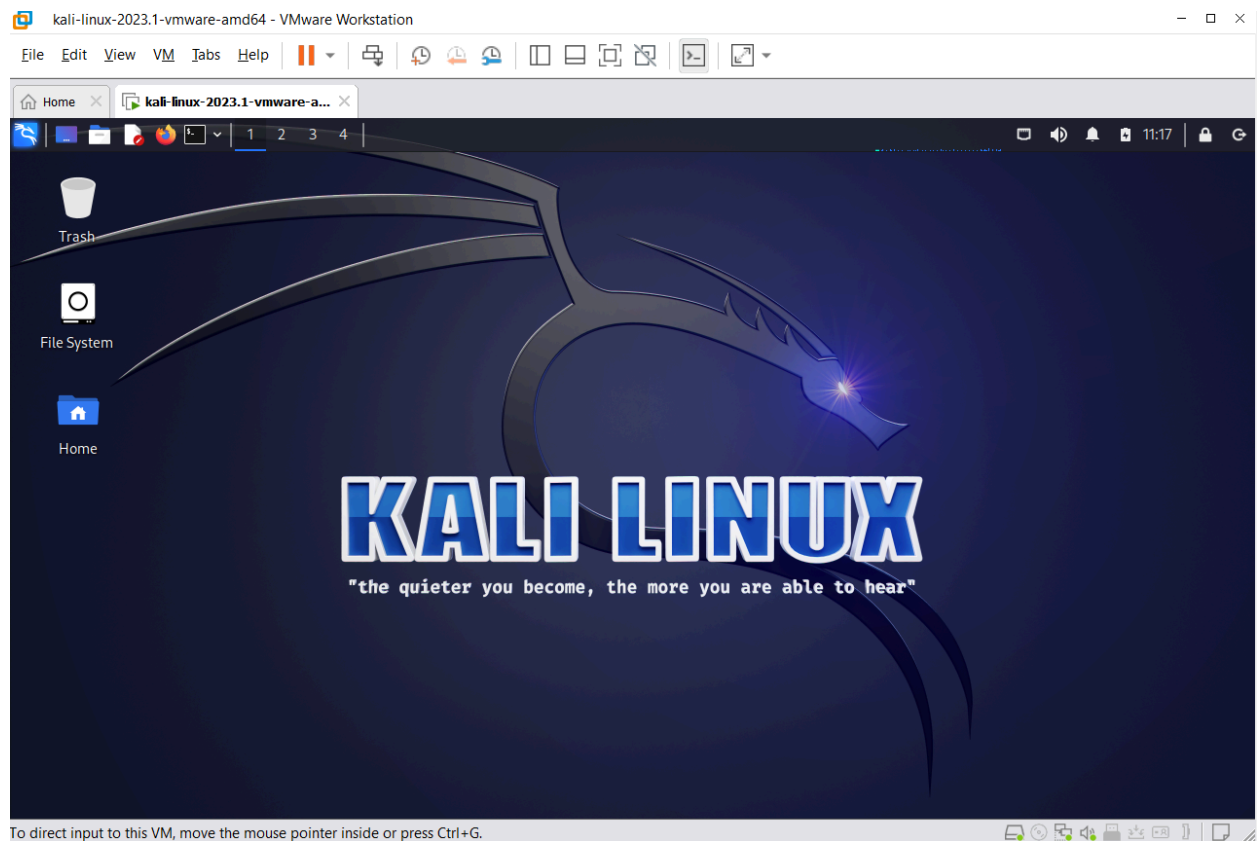


Locate the extracted Kali VMware virtual configuration file and click Open.



Click on “Power on this virtual machine” and Kali will load.



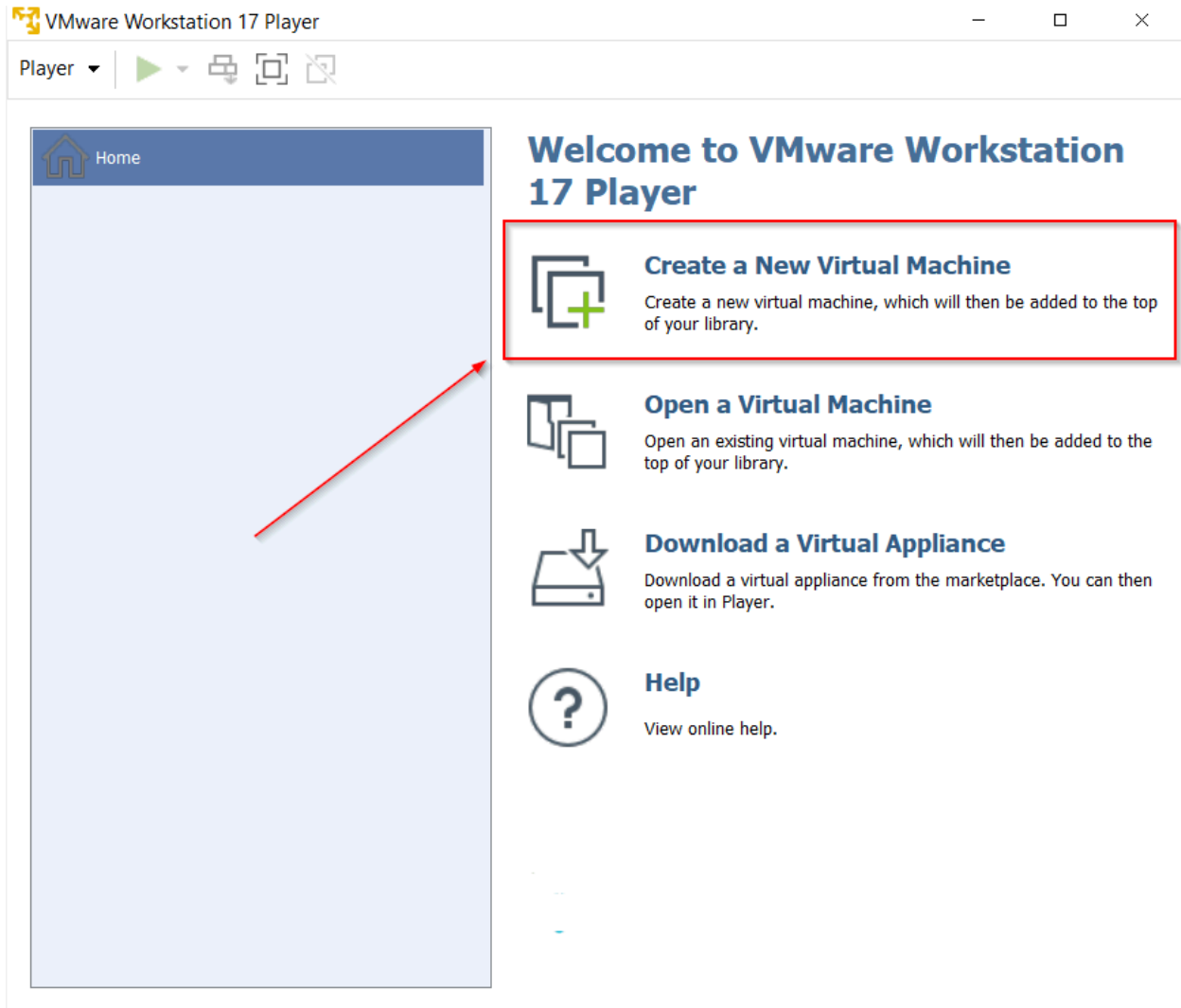


Installing Kali Linux From an ISO File

Installing the Kali Linux .iso file on VMware is identical whether you use the free Workstation Player or the paid Pro version. The only difference is that the Pro version allows you to customize the installation with advanced options such as SCSI controller type, virtual disk type, and compatibility with older VMware products.

Unless you absolutely need to use these options, we suggest installing Kali on Pro using the typical install.

Open either version of VMware and select “Create a New Virtual Machine.”



Browse to the location where the .iso is saved, then click Next.

New Virtual Machine Wizard ✕

Welcome to the New Virtual Machine Wizard

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

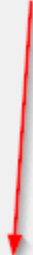
C:\Users\jupit\Downloads\Kali\kali-linux-2023.1-insta ▼ Browse...

 Could not detect which operating system is in this disc image.
You will need to specify which operating system will be installed.

☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel



In the dialogue box, you will be prompted to choose the Guest Operating System. Please select Linux as the Guest operating system and Debian 10.x 64-bit as the Version.

New Virtual Machine Wizard ✕

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows
☒ Linux
☐ Other

Version

Debian 10.x 64-bit ▼

Help < Back Next > Cancel

Next, set a name for your virtual machine and decide what folder you would like to use as the location of your virtual machine. You can choose whatever you would like in this step.

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:
Kali

Location:
C:\Users\jupit\Downloads\Kali [Browse...](#)

< Back Next > Cancel

Next, you will be asked to allocate disk space for your Kali virtual machine. Kali calls for 20 GB minimum, but if you have room suggest setting this to about 35 GB.

Select the option to Split the Virtual Disk into multiple files, which is the default choice. By doing this, the 20 to 35 GB you specify will not be used or reserved immediately.

Instead, these Virtual Disks will expand based on usage, up to the maximum size you've set as the disk capacity.

With a new installation, it typically takes up 10 GB of space, and this will grow as you install additional software within Kali.

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 20.0

Recommended size for Debian 10.x 64-bit: 20 GB

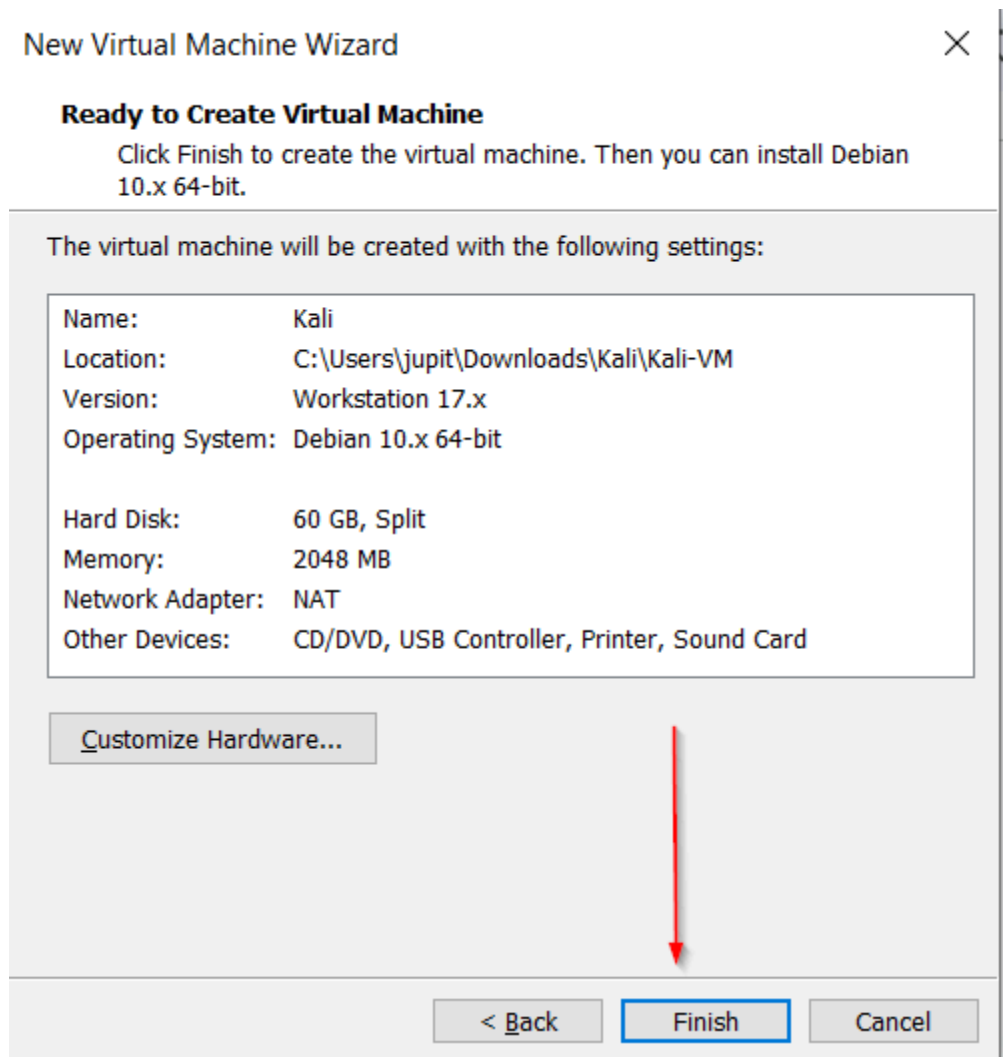
☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

At this point, you can click finish to complete the installation.



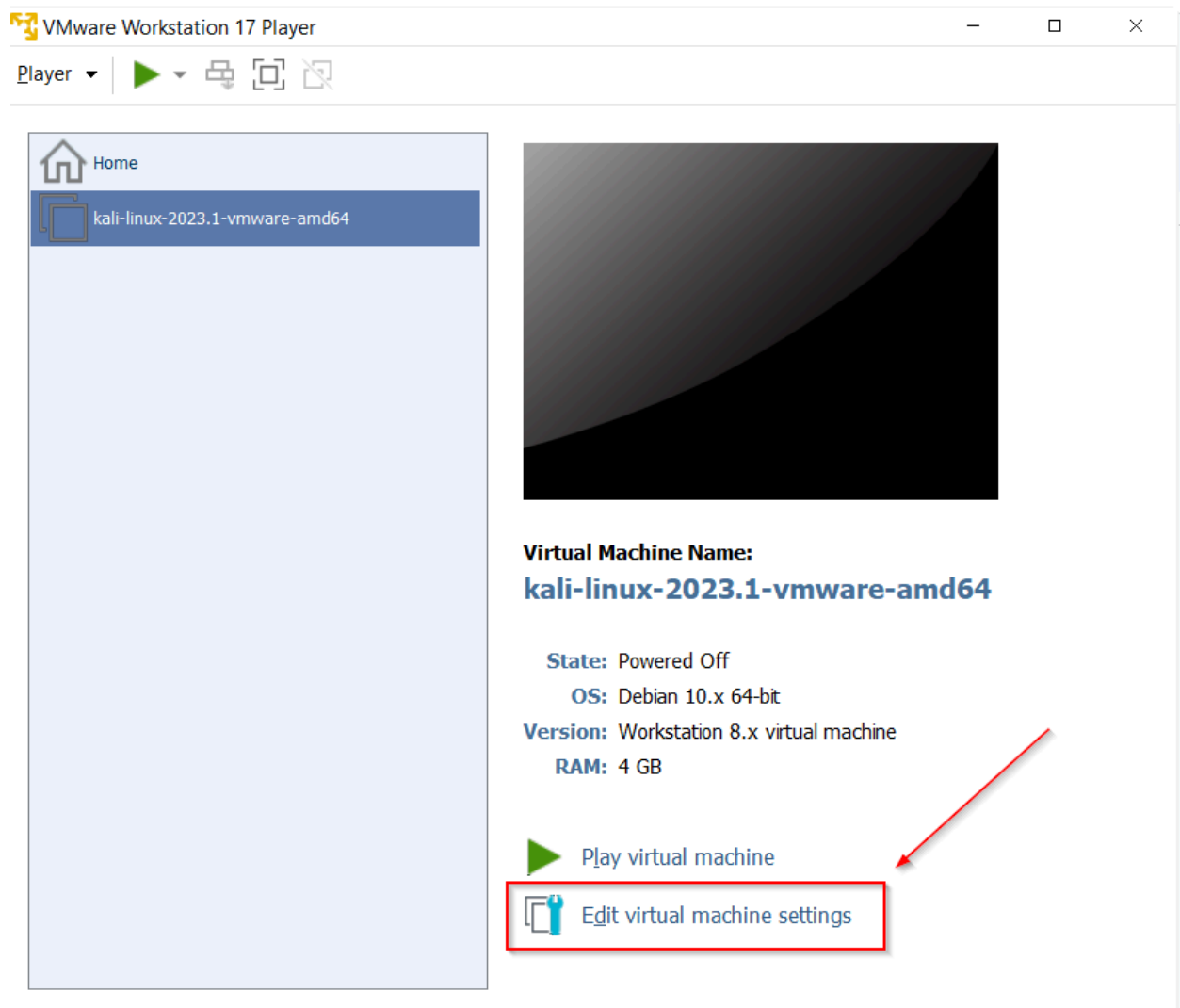
Once booted, the installation of Kali is identical whether you are installing it directly to your hardware (bare metal) or through a hypervisor like VMware. For a step-by-step guide on installing Kali once the iso has been set up,

Performance and Setup

VMware allows you to adjust many settings in your virtual machine, such as RAM allocation, processors used, taking snapshots, making a clone of a VM, and changing network configurations.

However, some features are only available in VMware Pro, including the snapshot feature and VM cloning, as well as the ability to change network configurations.

Changing the RAM and processors are features available in either setup. To access the settings menu in either edition, you need to click on "Edit virtual machine settings" from the main screen of VMware.

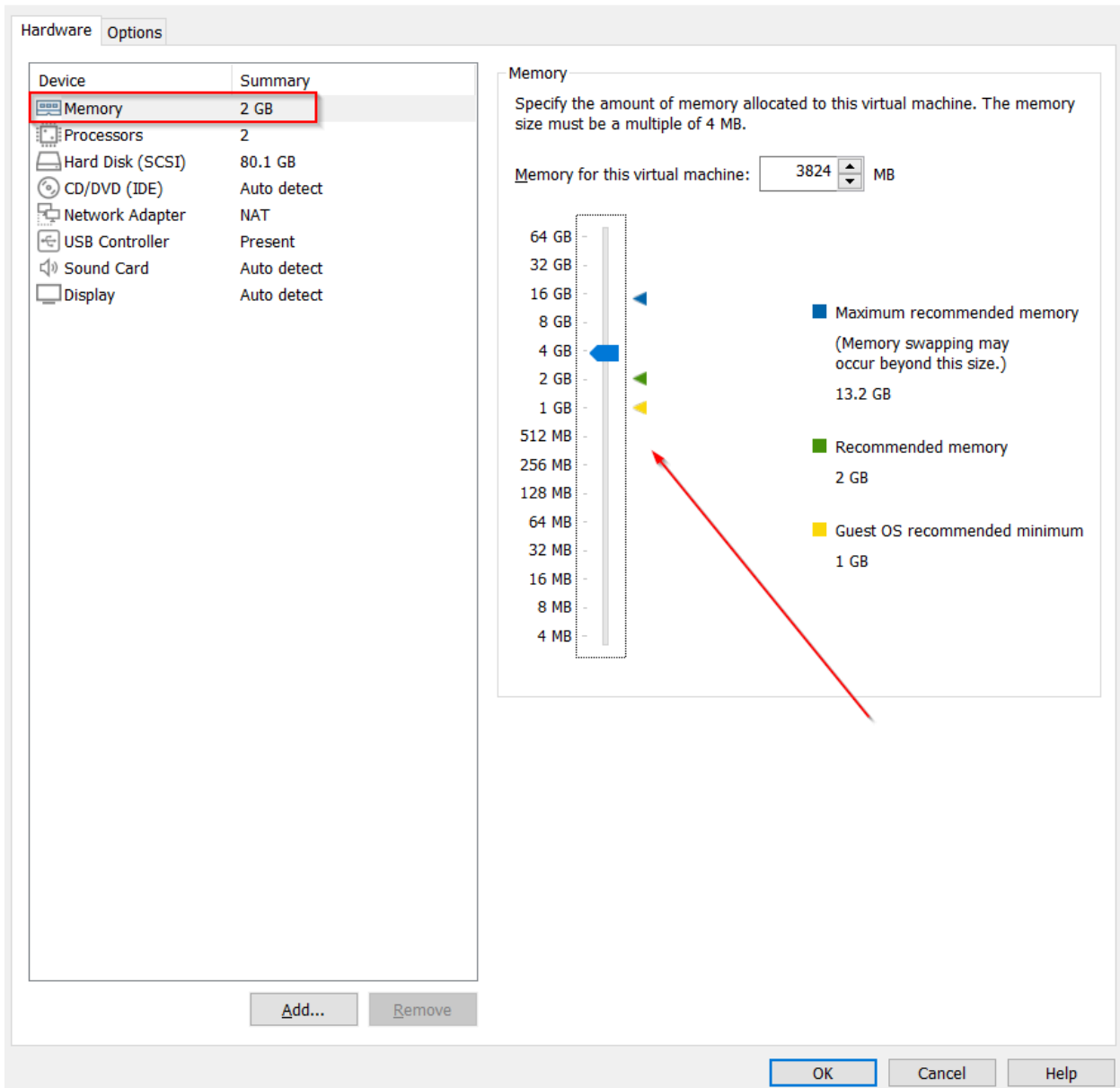


Increase RAM Allocation

To allocate more RAM to Kali Linux, click the "Memory" option and use the blue slider to adjust the size.

VMware provides a recommendation for the optimal RAM allocation. The slider allows you to assign anywhere from 4 MB to a maximum of 64 GB (assuming your host system has that much).

While we suggest allocating 4 GB of RAM for a smooth experience, Kali Linux can still function effectively with just 2 GB.



Creating Snapshots

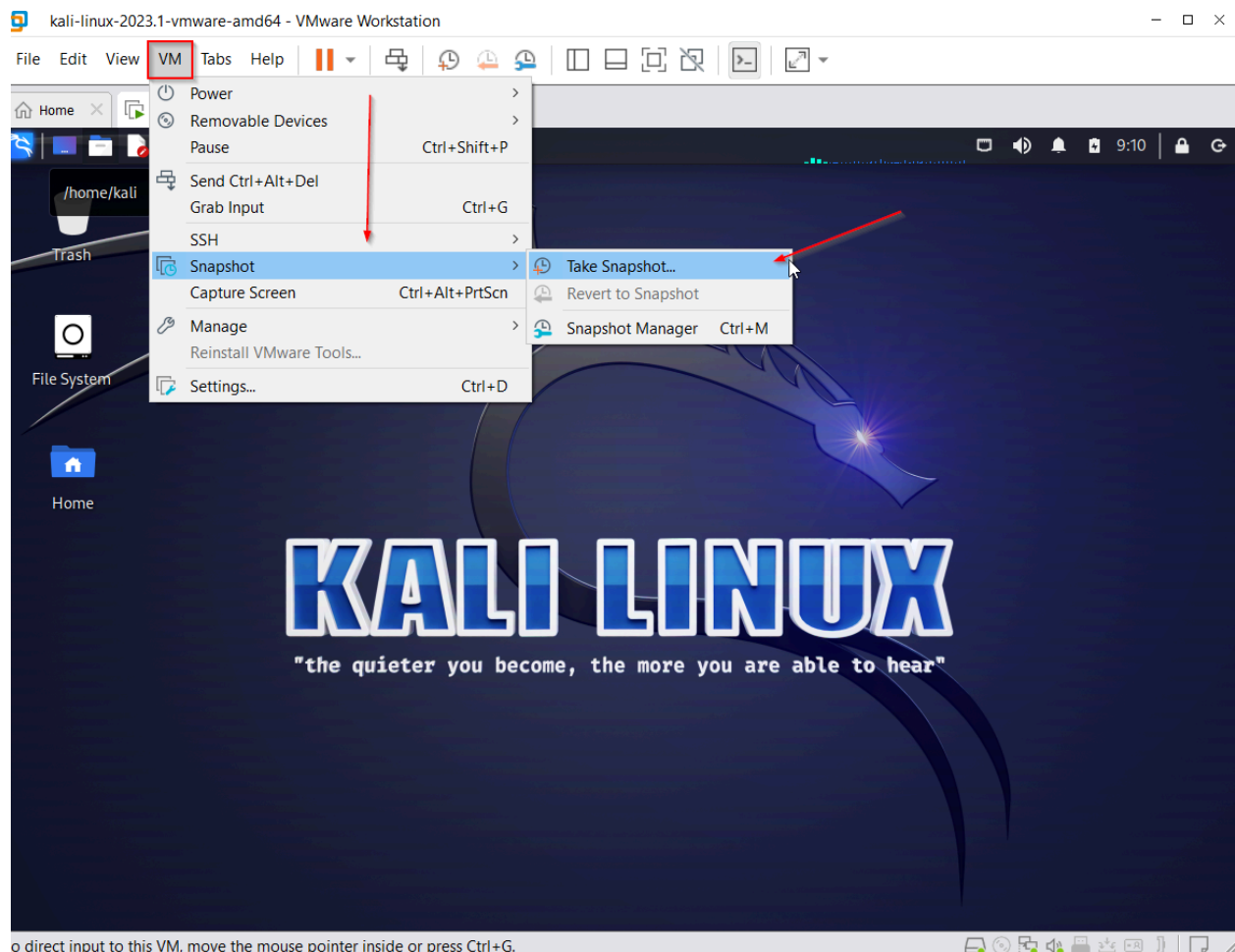
One useful feature in VMware, although limited to the paid version, is the ability to create snapshots.

A snapshot is essentially a way to capture the current state of your VM. It captures all of your settings and data at a point in time. This allows you to revert back to this saved state if anything goes wrong.

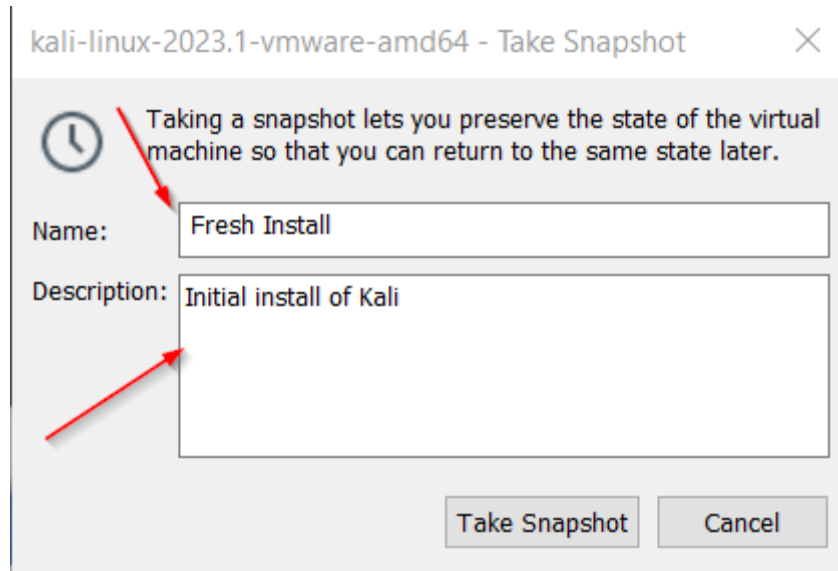
If you are using Workstation Pro, we recommend creating a snapshot as soon as you finish your initial installation of Kali. You can call it something like “fresh install.”

Going forward, we recommend creating a snapshot before any significant changes, such as updates or system configuration changes.

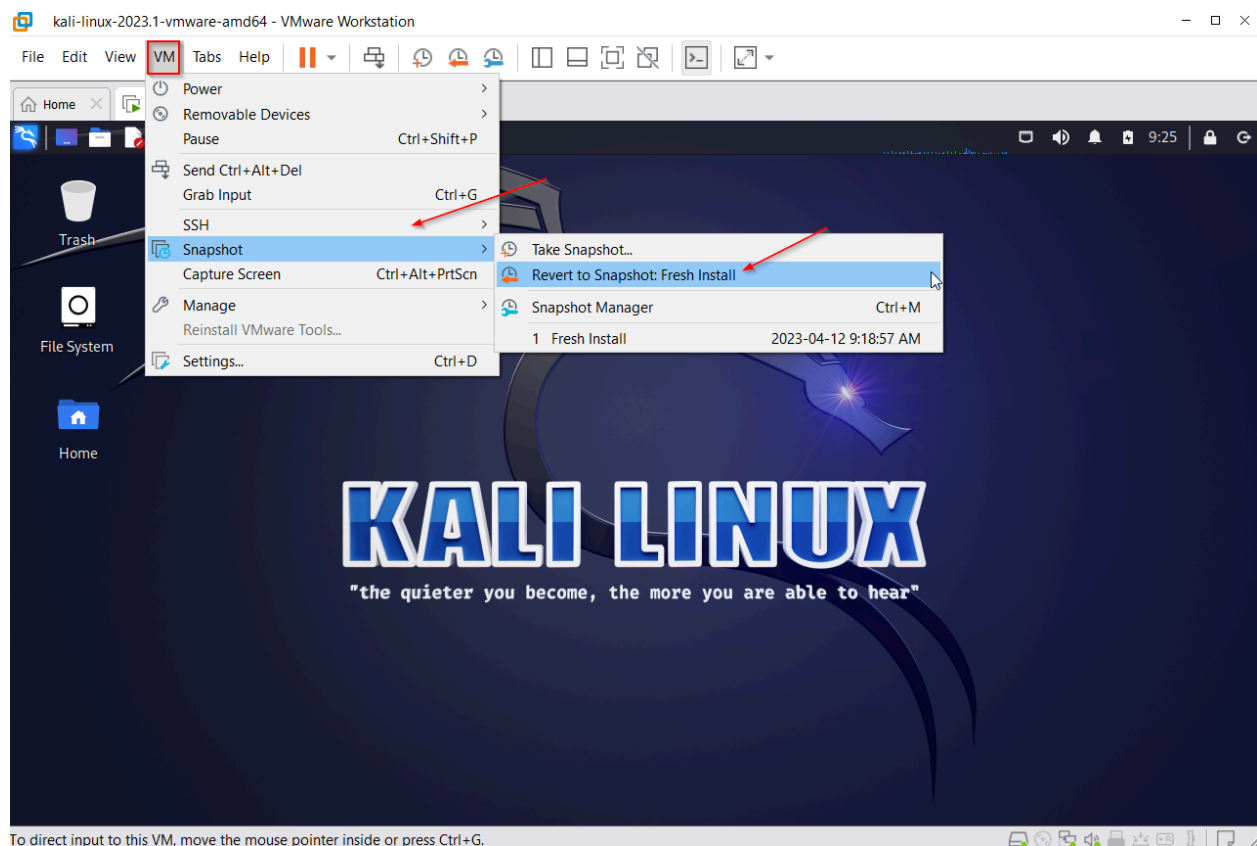
Creating a snapshot in Workstation Pro is straightforward. Head to the VM menu, select Snapshot, and then “Take Snapshot.”



Give your snapshot a name referencing why you are taking it, and give it a short description.



If you need to revert back to your snapshot. Head to the VM menu, select Snapshot, and then select “Revert to Snapshot.”



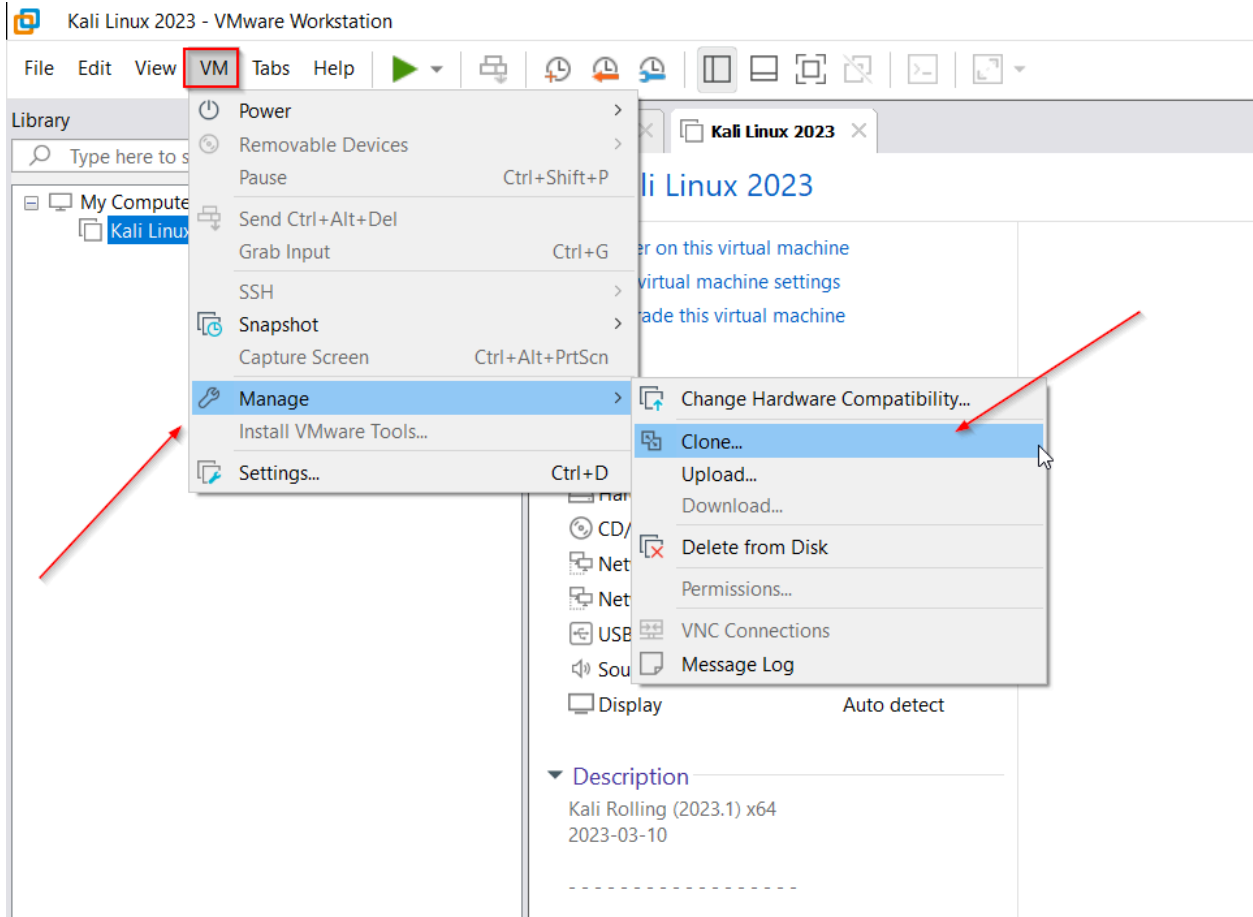
Cloning a VM

Another great feature included in VMware Pro is the ability to clone a VM. Now why would you want to do this? Put simply, it allows you to create numerous base systems where you can then test out different configurations.

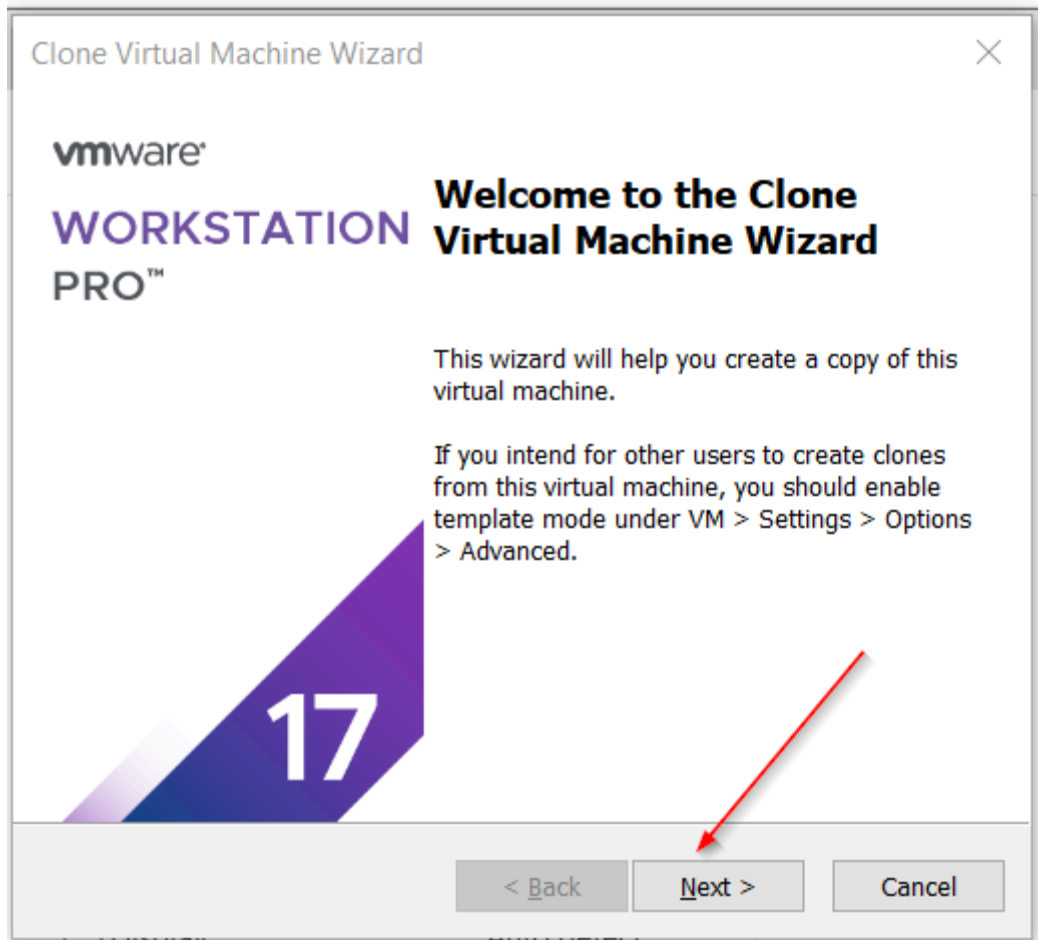
Unlike VM snapshots, where the original and new states cannot function at the same time, VM clones enable you to run multiple clones simultaneously.

Cloning will take up more disk space as it creates another VM, so please make sure you have sufficient space if you plan on using this feature.

To create a clone in VMware Pro, open the VM menu, next click on Manage, and finally click on “Clone.”



Click Next to start the Clone Virtual Machine Wizard.



Choose where you want to clone from and click Next.

Clone Virtual Machine Wizard ×

Clone Source
Which state do you want to create a clone from?

Clone from

☒ The current state in the virtual machine
Creating a linked clone from the current state will create a new snapshot.

☐ An existing snapshot (powered off only):

This virtual machine has no existing cloneable snapshots.

< Back Next > Cancel

Choose “Create a full clone” (recommended) and click Next.

Clone Virtual Machine Wizard ✕

Clone Type
How do you want to clone this virtual machine?

Clone method

☐ Create a linked clone
A linked clone is a reference to the original virtual machine and requires less disk space to store. However, it cannot run without access to the original virtual machine.

☒ Create a full clone 
A full clone is a complete copy of the original virtual machine at its current state. This virtual machine is fully independent, but requires more disk space to store.

< Back Next > Cancel

Finally, name the new clone and decide where you would like it to be saved.

Clone Virtual Machine Wizard

✕

Name of the New Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name

Clone of Kali Linux 2023

Location

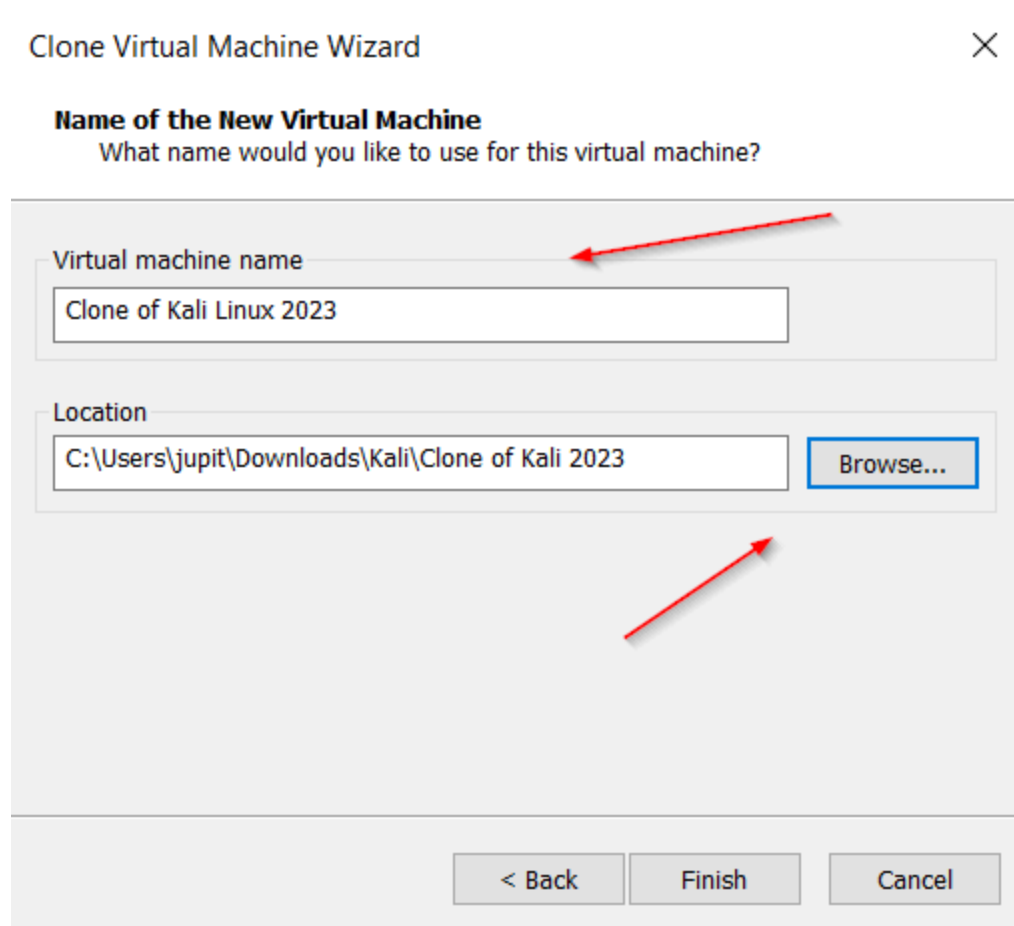
C:\Users\jupit\Downloads\Kali\Clone of Kali 2023

Browse...

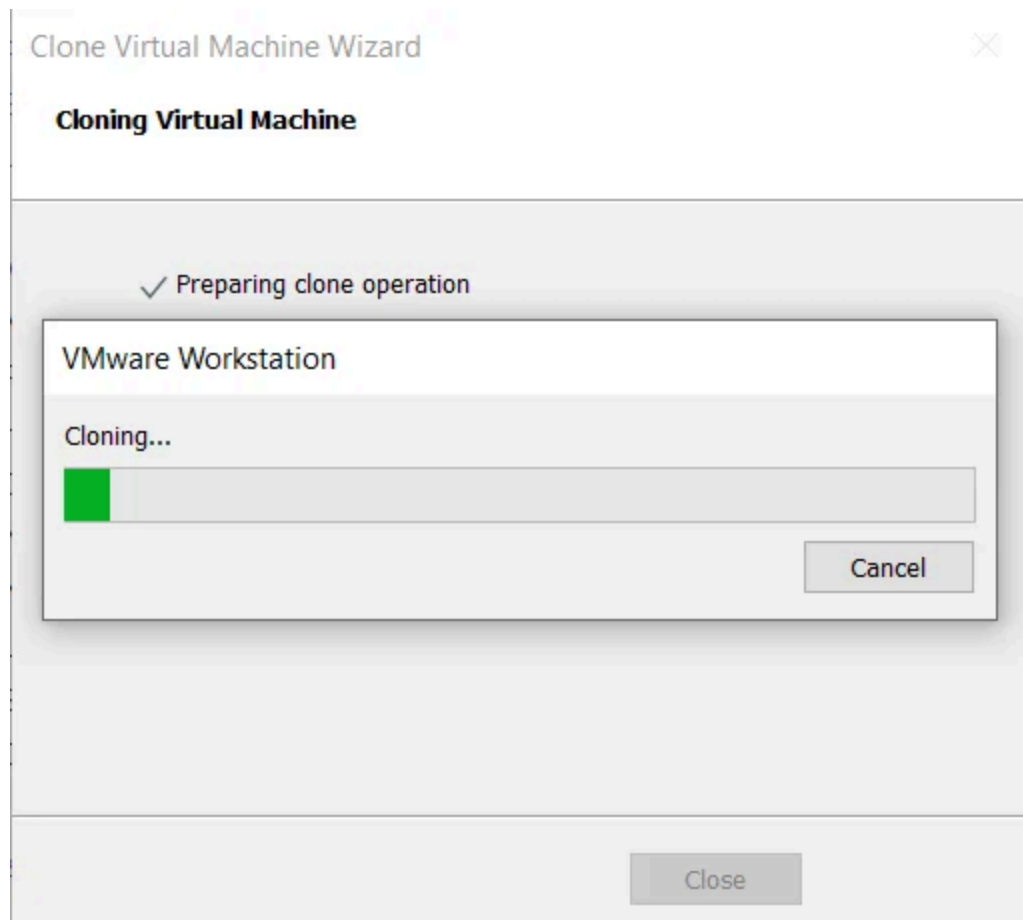
< Back

Finish

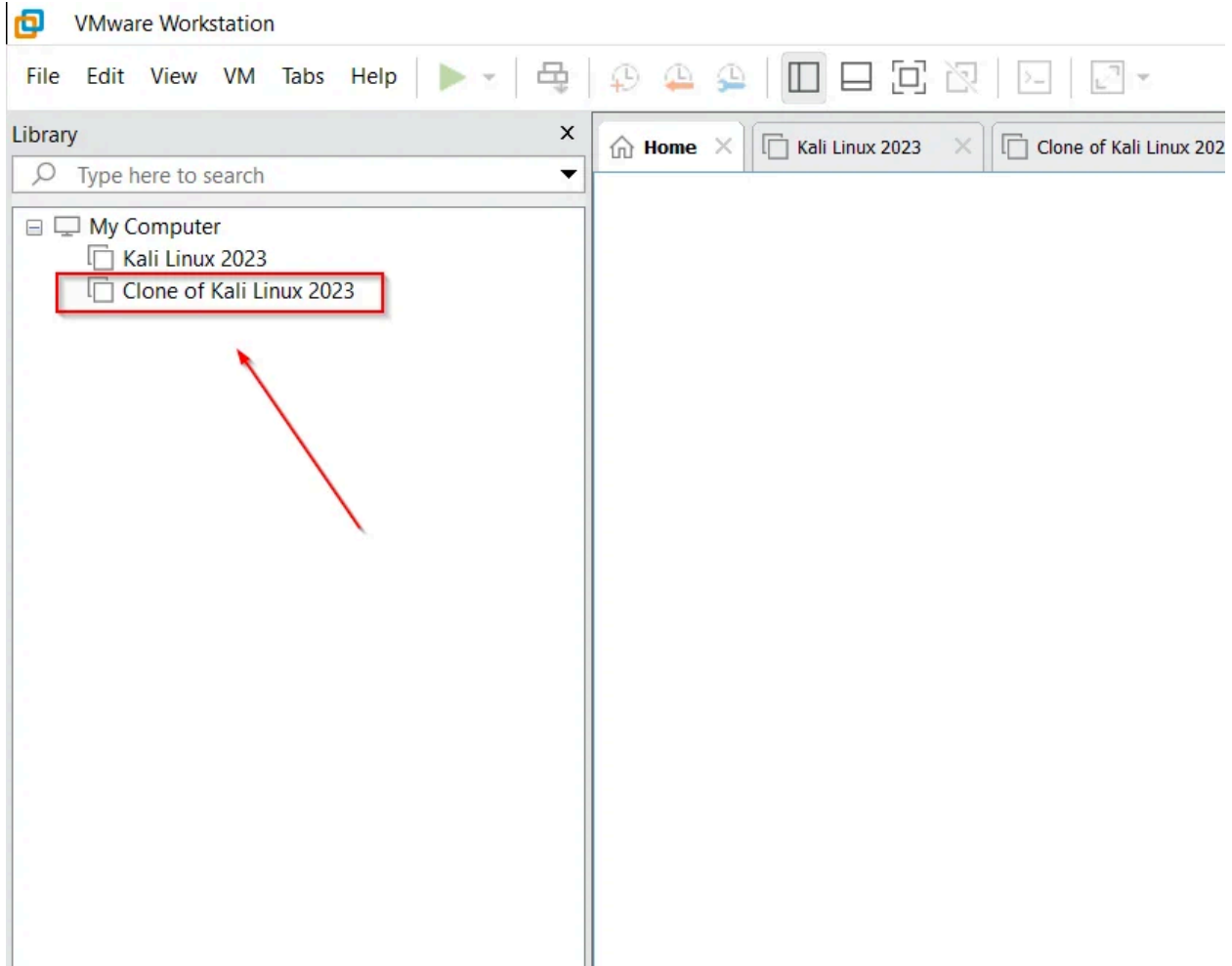
Cancel



The cloning process will take a few minutes to complete.



You now have a cloned VM.

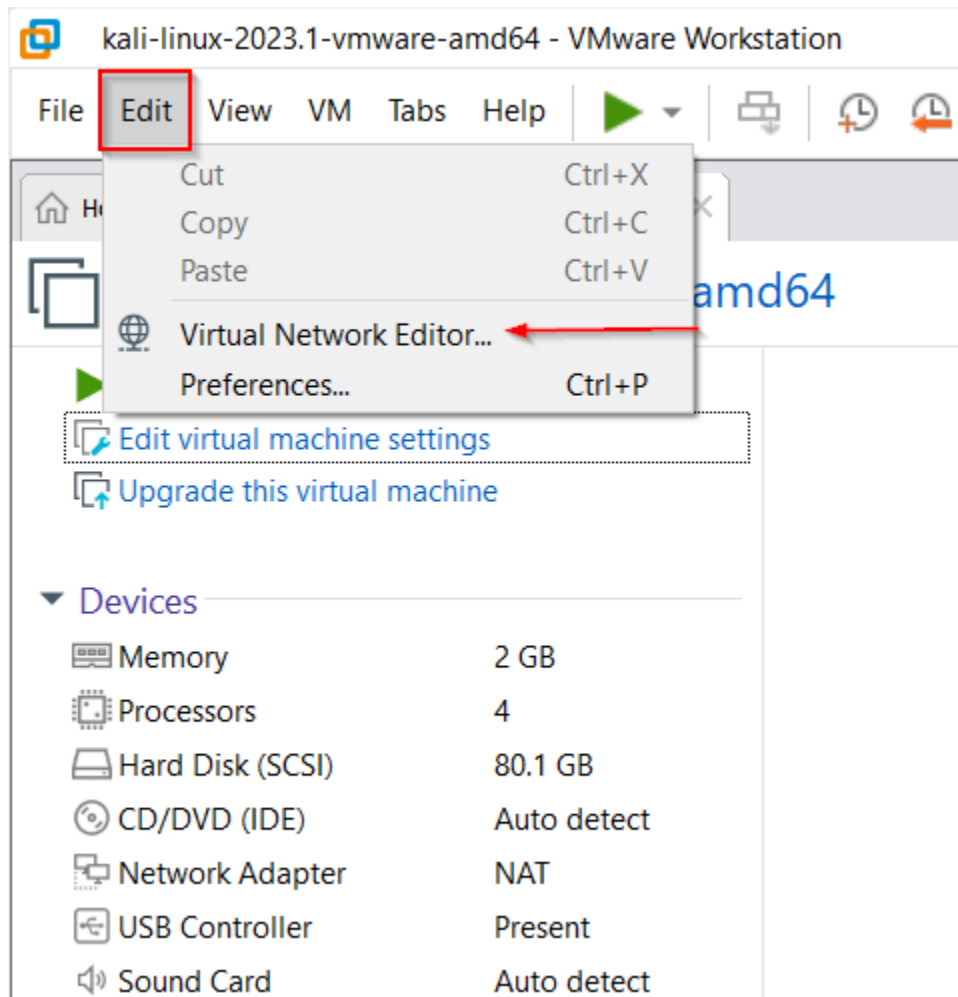


Change Network Settings

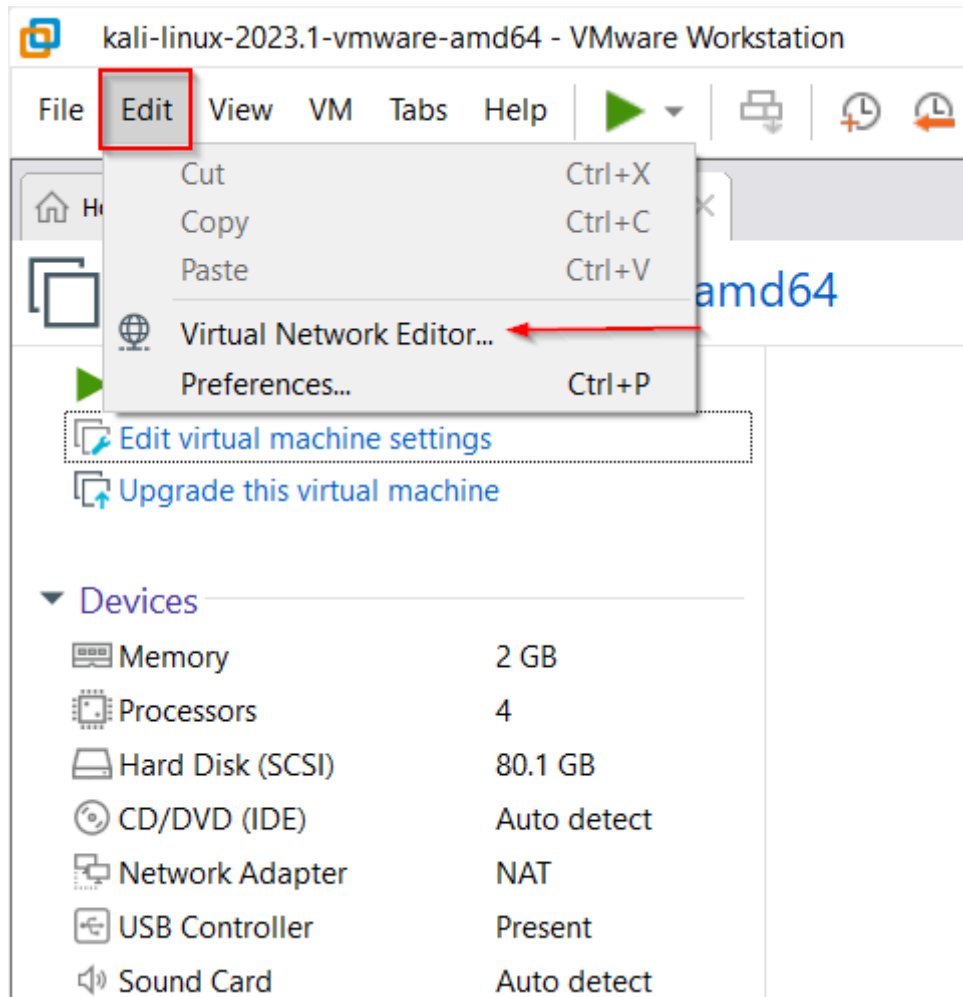
VMware allows you to change Network Settings. However, you are fairly limited in the free Player version, as you can only change whether you use NAT, host, or bridged mode.

In the Pro version, you have the same options as well as the added benefit of creating new virtual networks, which can be useful if you are

running a lab with numerous machines and you want certain ones on a separate network, such as when practicing pivoting.



To create a new network, select the Edit menu and then “Virtual Network Editor.”



Select Change settings. You will need Administrator privileges to make these changes.



Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet1	Host-only	-	Connected	Enabled	192.168.87.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.110.0

[Add Network...](#) [Remove Network](#) [Rename Network...](#)

VMnet Information
☐ Bridged (connect VMs directly to the external network)
Bridged to: [Automatic Settings...](#)
☐ NAT (shared host's IP address with VMs) [NAT Settings...](#)
☒ Host-only (connect VMs internally in a private network)

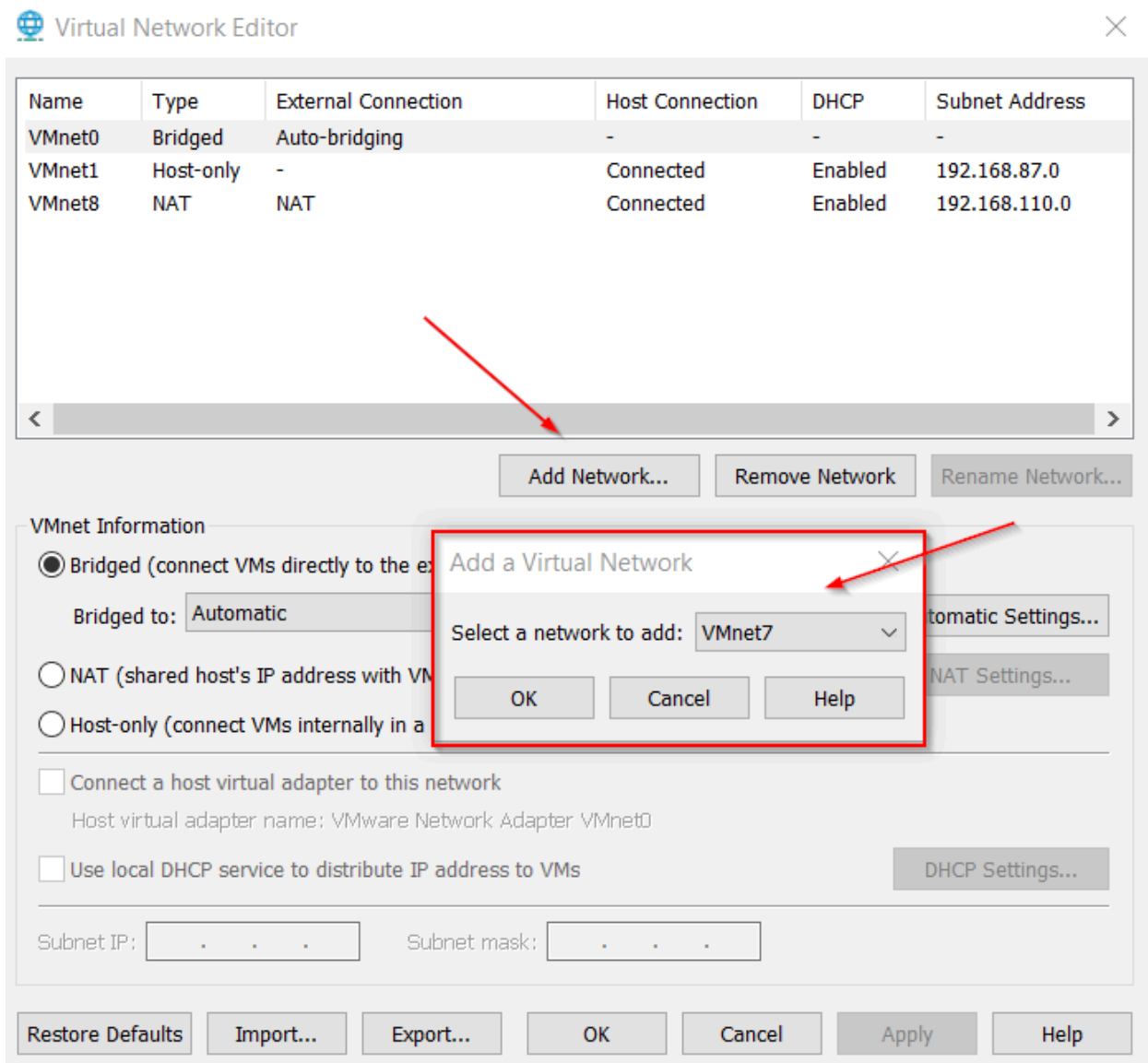
☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet1
☒ Use local DHCP service to distribute IP address to VMs [DHCP Settings...](#)

Subnet IP: Subnet mask:

Administrator privileges are required to modify the network configuration. [Change Settings](#)

[Restore Defaults](#) [Import...](#) [Export...](#) [OK](#) [Cancel](#) [Apply](#) [Help](#)

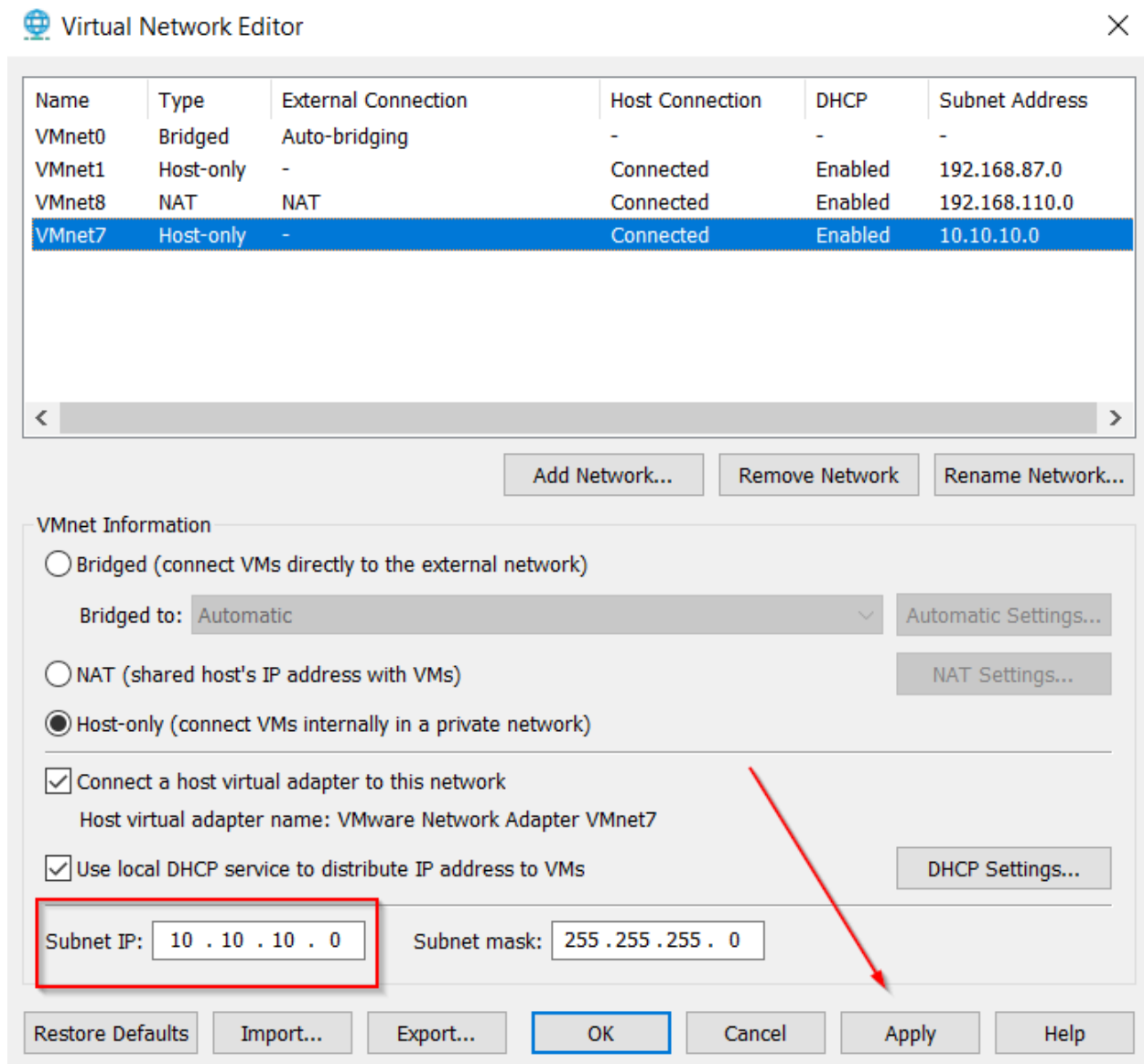
Click add network and select a network to add. We chose VMnet7, but you can choose any available network.



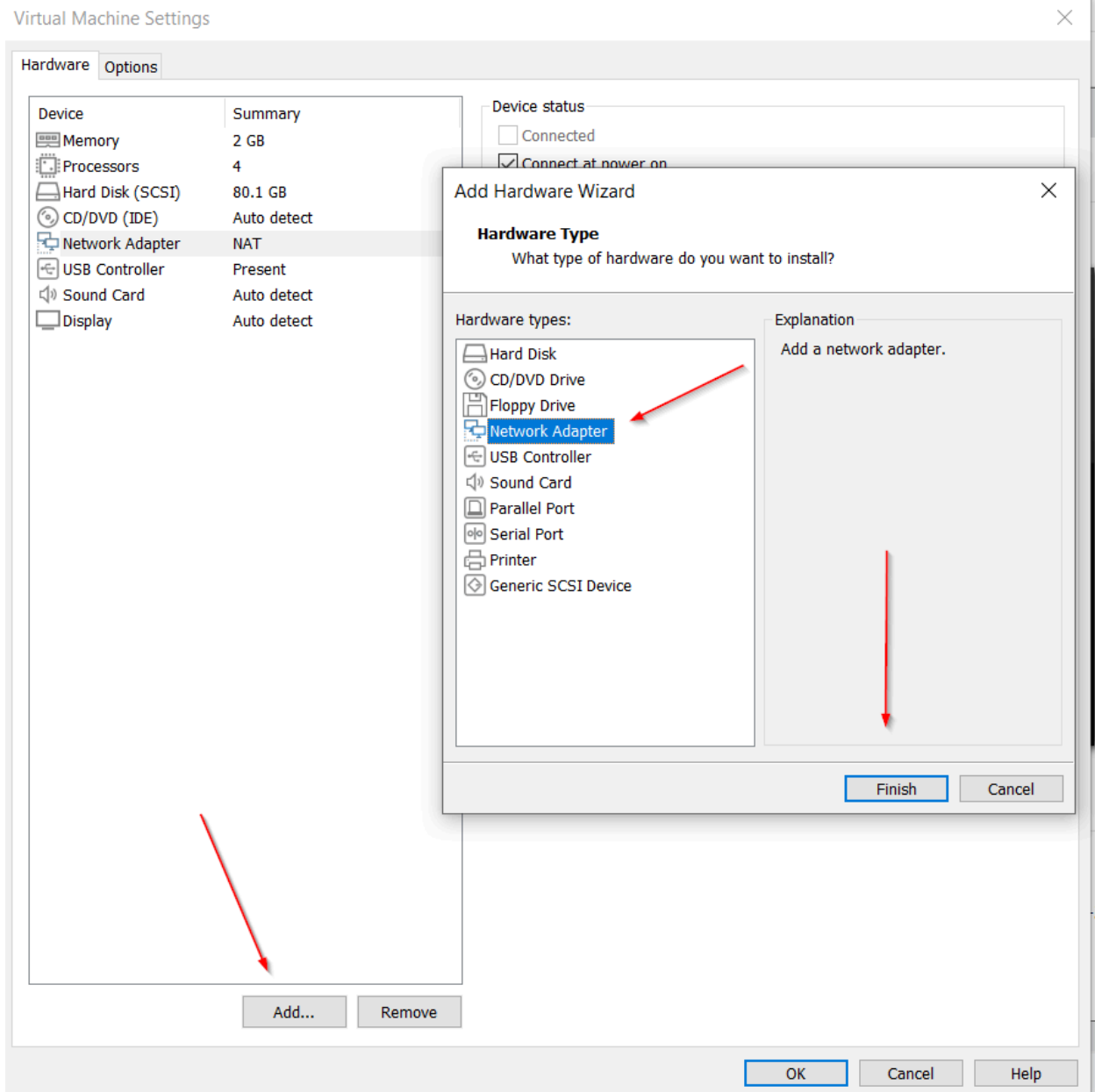
You can leave the subnet IP as is, but we changed ours to 10.10.10.0. If you change your subnet IP, ensure you have the correct subnet mask.

If you are looking for a subnetting cheatsheet, we have you covered. Check out our [**The Only IPv4 Subnetting Cheat Sheet You'll Ever Need**](#) blog post now.

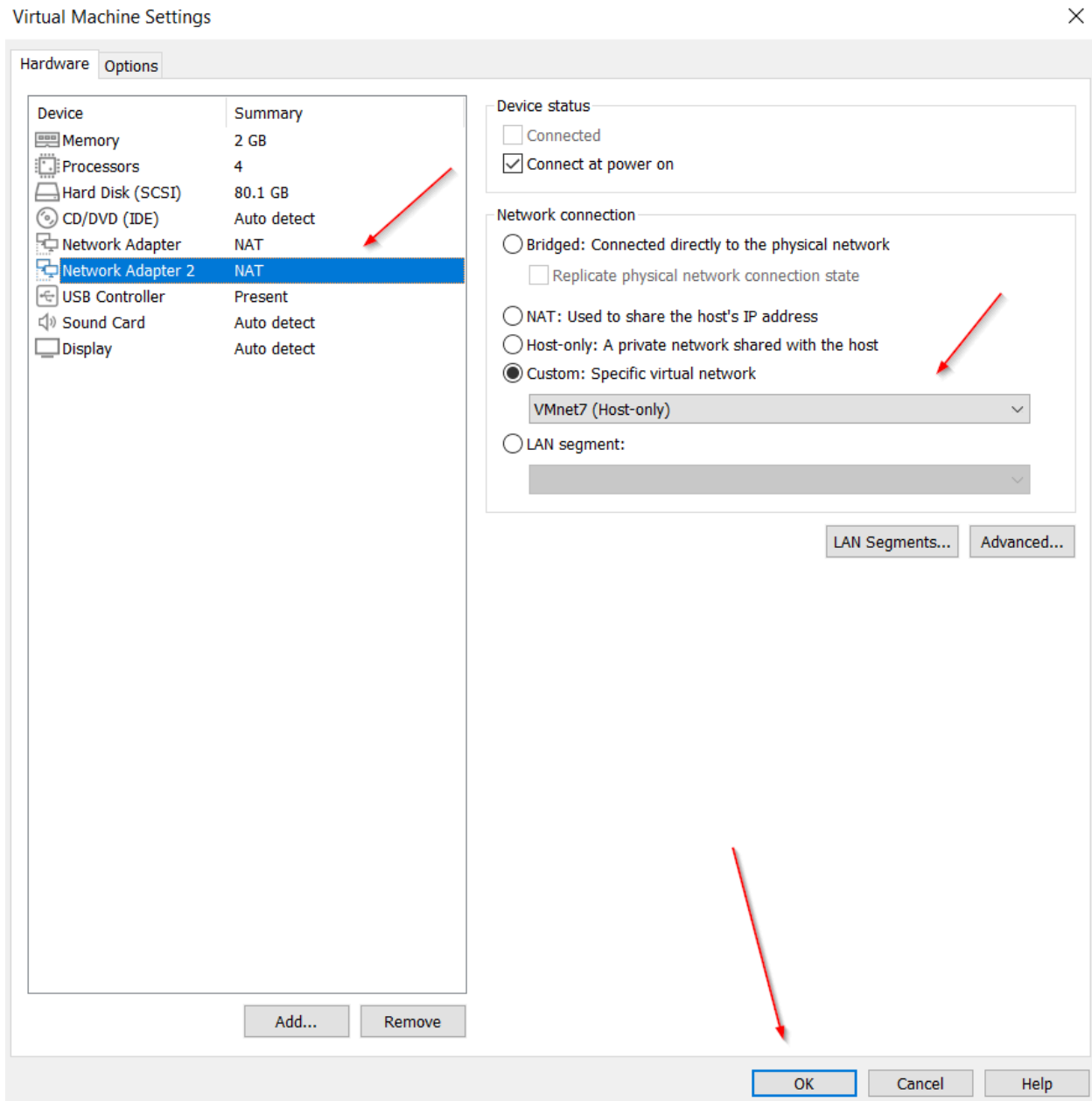
Select the VMnet7 network and change the subnet IP. Click Apply. It will take about a minute or two to create the network. Once completed, you will have a new subnet to which you can assign your Kali Linux in VMware.



To add our Kali Linux VM to the created VMnet7 network, we need to add a second network adapter. Select "Edit virtual machine settings", select add, select the network adapter, and finally, select Finish.



Once we have the new network adapter added to Kali, we can assign it to the new subnet we created.



You should now have Kali with two network adapters, also known as dual-homed, each connected to different subnetworks. You can confirm this by starting Kali and opening a terminal. In the terminal, type: ip a

```
(kali㉿kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:53:83:62 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.110.130/24 brd 192.168.110.255 scope global dynamic noprefixroute eth0  
        valid_lft 1771sec preferred_lft 1771sec  
    inet6 fe80::d589:faf3:ddee:a7c9/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:53:83:6c brd ff:ff:ff:ff:ff:ff  
    inet 10.10.10.128/24 brd 10.10.10.255 scope global dynamic noprefixroute eth1  
        valid_lft 1771sec preferred_lft 1771sec  
    inet6 fe80::fd00:527:5768:f70e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

Conclusion

As you've seen, VMware is a very efficient and powerful piece of software that provides a streamlined process and offers free and paid versions depending on your needs.

With our step-by-step guide on how to install Kali Linux on VMware, you should be well on your way to setting up and optimizing your virtual environment and start hacking. Just follow the instructions, and you'll have Kali Linux running smoothly on VMware in no time.